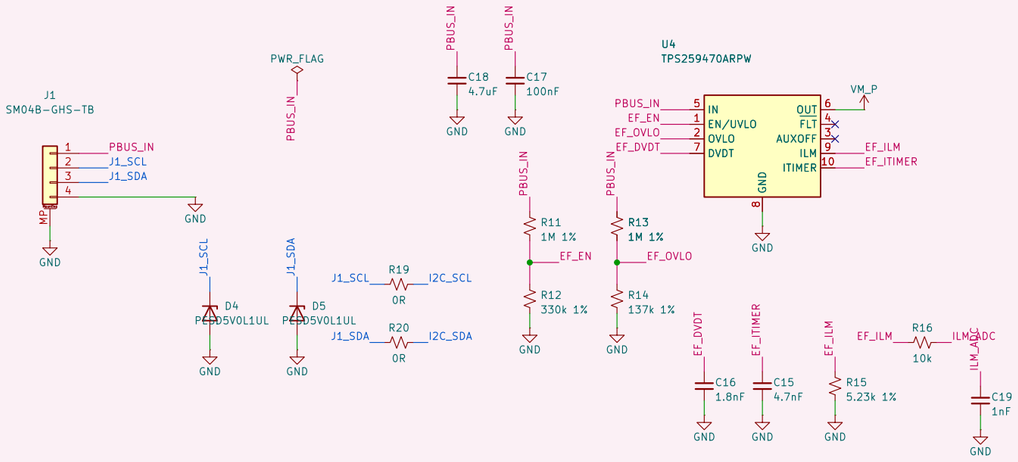
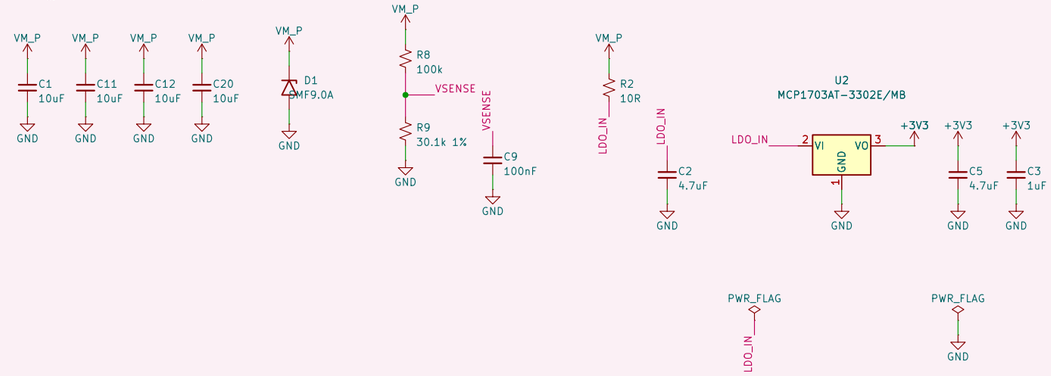
HOST J1 (PBUS v2 + I2C), eFUSE U4 TPS259470A



J1: 1 PBUS 2 SCL 3 SDA 4 GND, MP = GND. Silk "PBUS 6–9 V".
No I2C pull-ups here (host 4.7k). R19/R20 0R (<= 100R allowed).

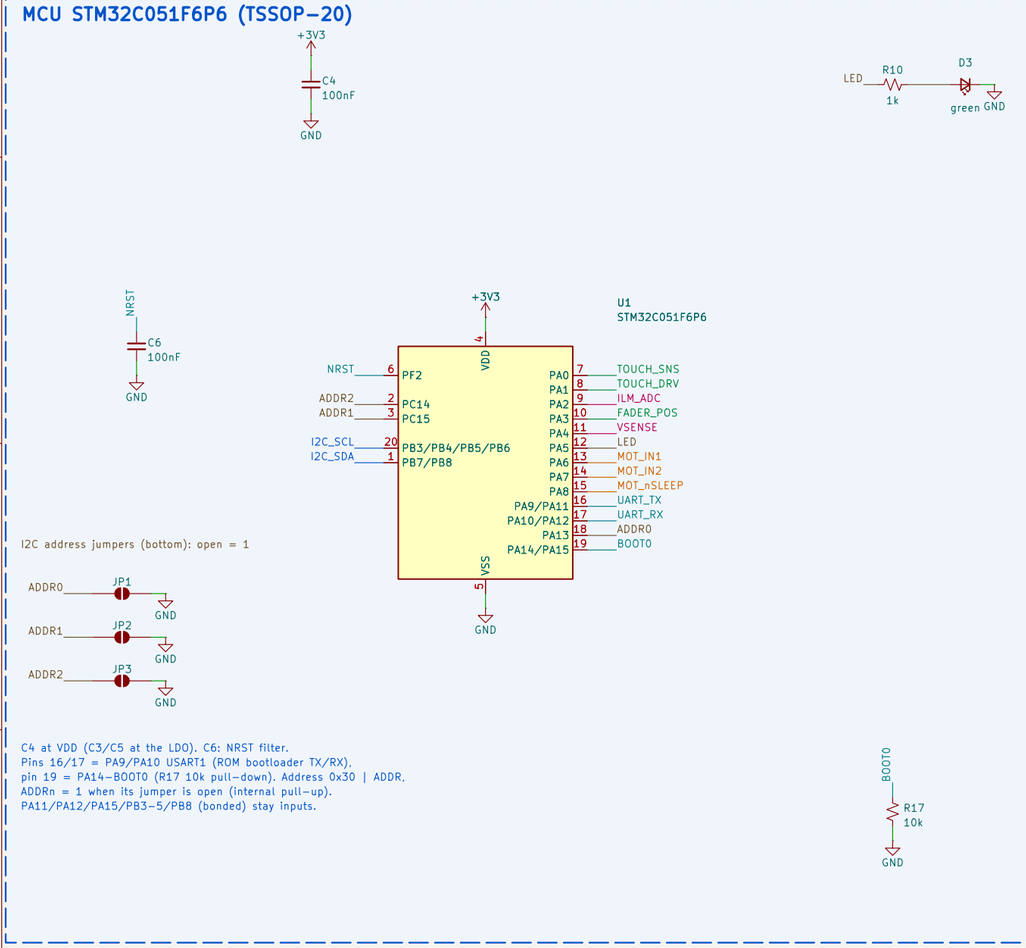
U4 TPS259470A (auto-retry): UVLO 4.84/4.40 V, OVLO 9.96/9.05 V, I_LIM 3334/5.23k = 0.64 A (0.54–0.73 A),
ITIMER 4.7 nF ~3.9 ms, dVdt 1.8 nF ~1.1 V/ms. Reverse blocking to −15 V, survives 11 V (OVLO off).
ILM monitor 0.95 V/A -> R16 10k + C19 1 nF -> PA2. FLT and AUXOFF unused.

VM_P (protected PBUS), PBUS SENSE, 3V3 LDO



VM_P bulk 4 x 10 uF 25 V (~45 uF total input incl. C18). D1 SMF9.0A clamps back-EMF on VM_P.
VSENSE = VM_P x 30.1/130.1 -> PA4 (11 V -> 2.55 V). U2 MCP1703A SOT-89 from VM_P via R2 10R.
J2 pin 2 (3V3) feeds LDO_IN through D7 BAT54W, so VOUT never exceeds VIN (MCP1703A abs max).

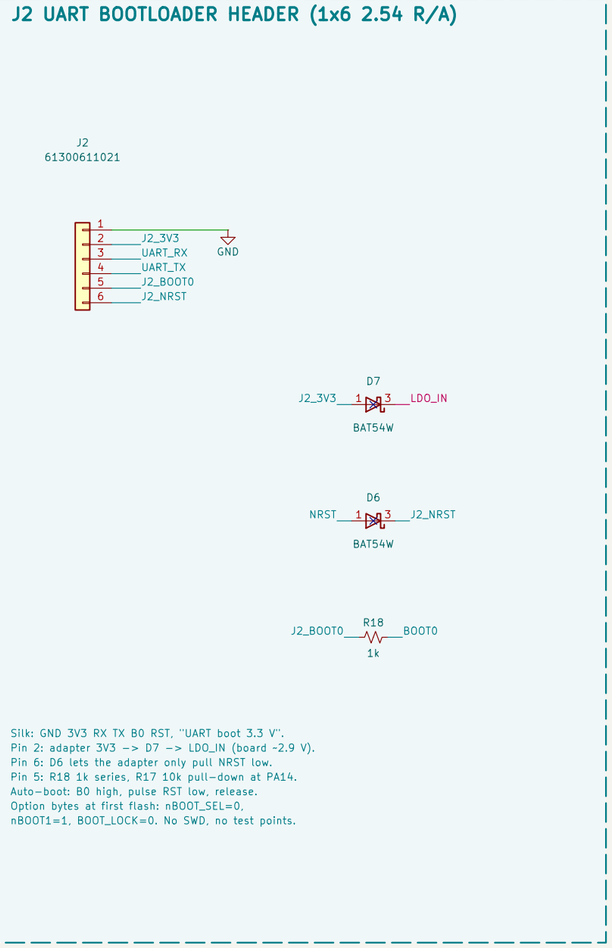
MCU STM32C051F6P6 (TSSOP-20)



I2C address jumpers (bottom): open = 1

C4 at VDD (C3/C5 at the LDO). C6: NRST filter.
Pins 16/17 = PA9/PA10 USART1 (ROM bootloader TX/RX),
pin 19 = PA14–BOOT0 (R17 10k pull-down). Address 0x30 | ADDRn,
ADDRn = 1 when its jumper is open (internal pull-up).
PA11/PA12/PA15/PB3–5/PB8 (bonded) stay inputs.

J2 UART BOOTLADER HEADER (1x6 2.54 R/A)

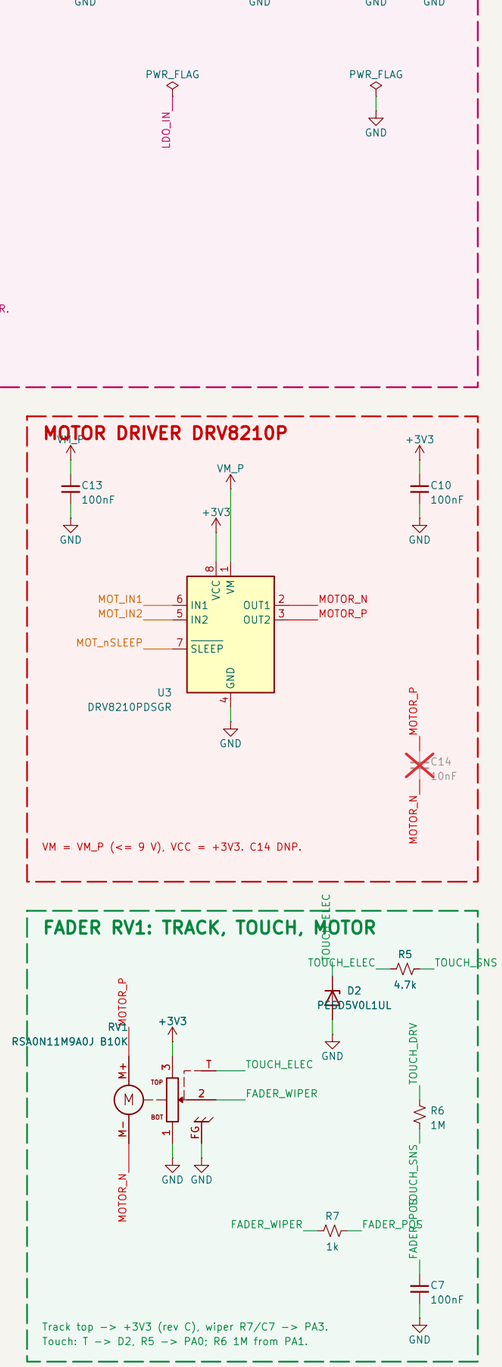


Silk: GND 3V3 RX TX B0 RST, "UART boot 3.3 V".
Pin 2: adapter 3V3 -> D7 -> LDO_IN (board ~2.9 V).
Pin 6: D6 lets the adapter only pull NRST low.
Pin 5: R18 1k series, R17 10k pull-down at PA14.
Auto-boot: B0 high, pulse RST low, release.
Option bytes at first flash: nBOOT_SEL=0,
nBOOT1=1, BOOT_LOCK=0. No SWD, no test points.

MOTOR DRIVER DRV8210P

VM = VM_P (<= 9 V), VCC = +3V3. C14 DNP.

FADER RV1: TRACK, TOUCH, MOTOR



Track top -> +3V3 (rev C), wiper R7/C7 -> PA3.
Touch: T -> D2, R5 -> PA0; R6 1M from PA1.

Spec docs/SPEC_revC.md v1.0; generated by build/scripts/gen_sch.py
PBUS v2 (6–9 V) + I2C on JST GH 4p; eFuse TPS259470A; UART bootloader header J2

Sheet: /
File: E0001_fader_driver.kicad_sch

Title: E0001 Motorized fader driver board

Size: User	Date: 2026-10-19	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 1/1